SmartMatch

A Multi-Agent Pipeline for Mood Board Generation via Graph-Augmented Retrieval and Multimodal Synthesis

Team 4 - Caroline Delva · Xinzhou Li · Nandini Kodali · Qingyang Wang

GEORGETOWN UNIVERSITY

Scan Here to Try !

INTRODUCTION

Creating a mood board requires translating emotional intent into coherent visual compositions — a task where conventional image retrieval falls short, as abstract and affective language does not map naturally to the visual feature spaces of retrieval models. SmartMatch addresses this challenge through a multi-agent pipeline that decomposes user intent into structured visual descriptors, retrieves candidates via hybrid SigLIP-2 and text embeddings over 25K+ Unsplash images, re-ranks and expands results through a graph-augmented knowledge structure, and verifies cross-image coherence before generating explainable justifications.

DATA OVERVIEW

- The image corpus consists of 25,000 photographs from the Unsplash dataset.
- Grounding outputs for all 25,000 images were generated offline via the Anthropic Batch API.
- SigLIP-2 embeddings are stored in a FAISS flat inner product index with L2 normalization.
- Field text embeddings produce a 25,000 x 3,072 matrix per field stored as NumPy float32 arrays.

Flowchart

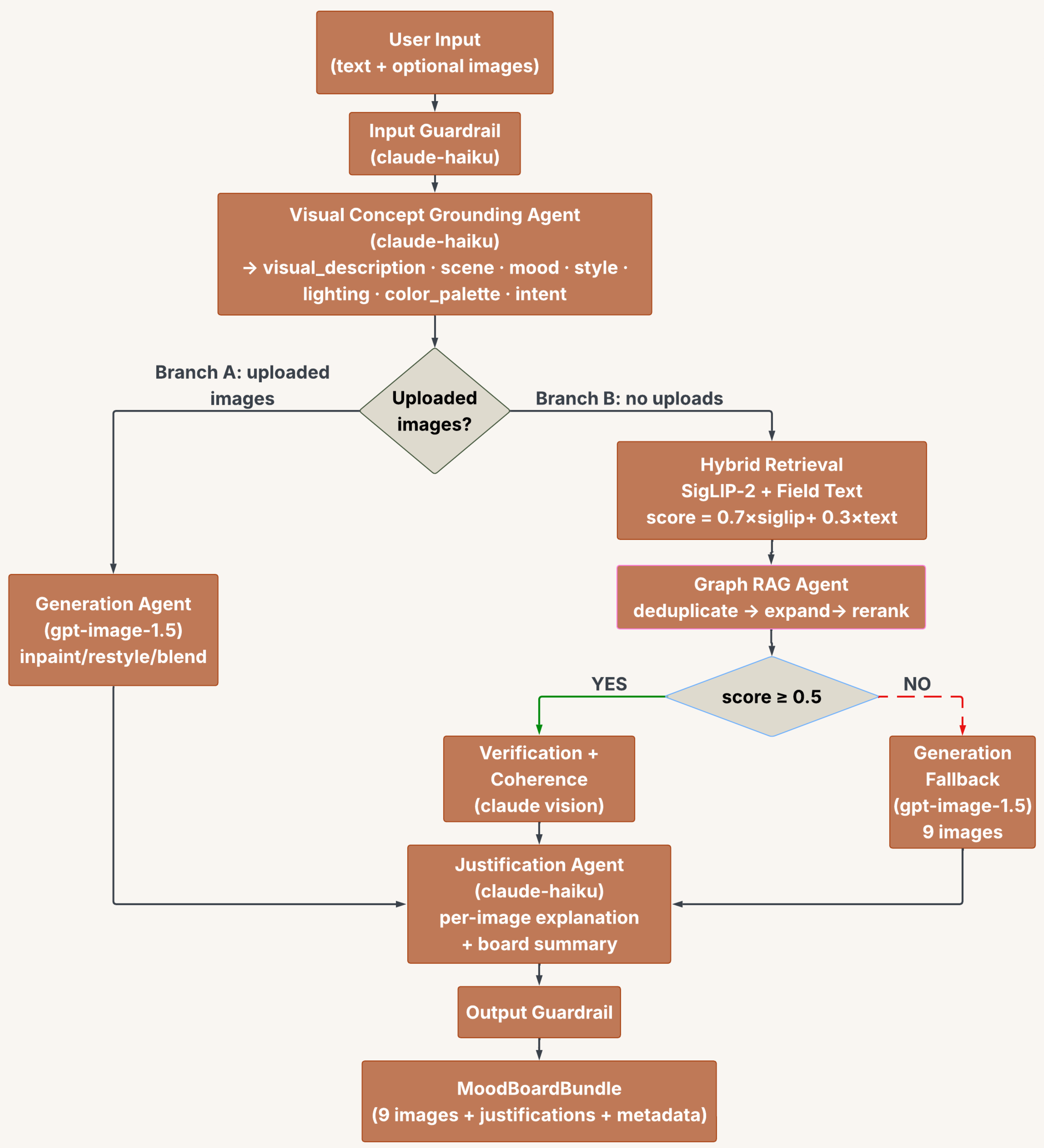
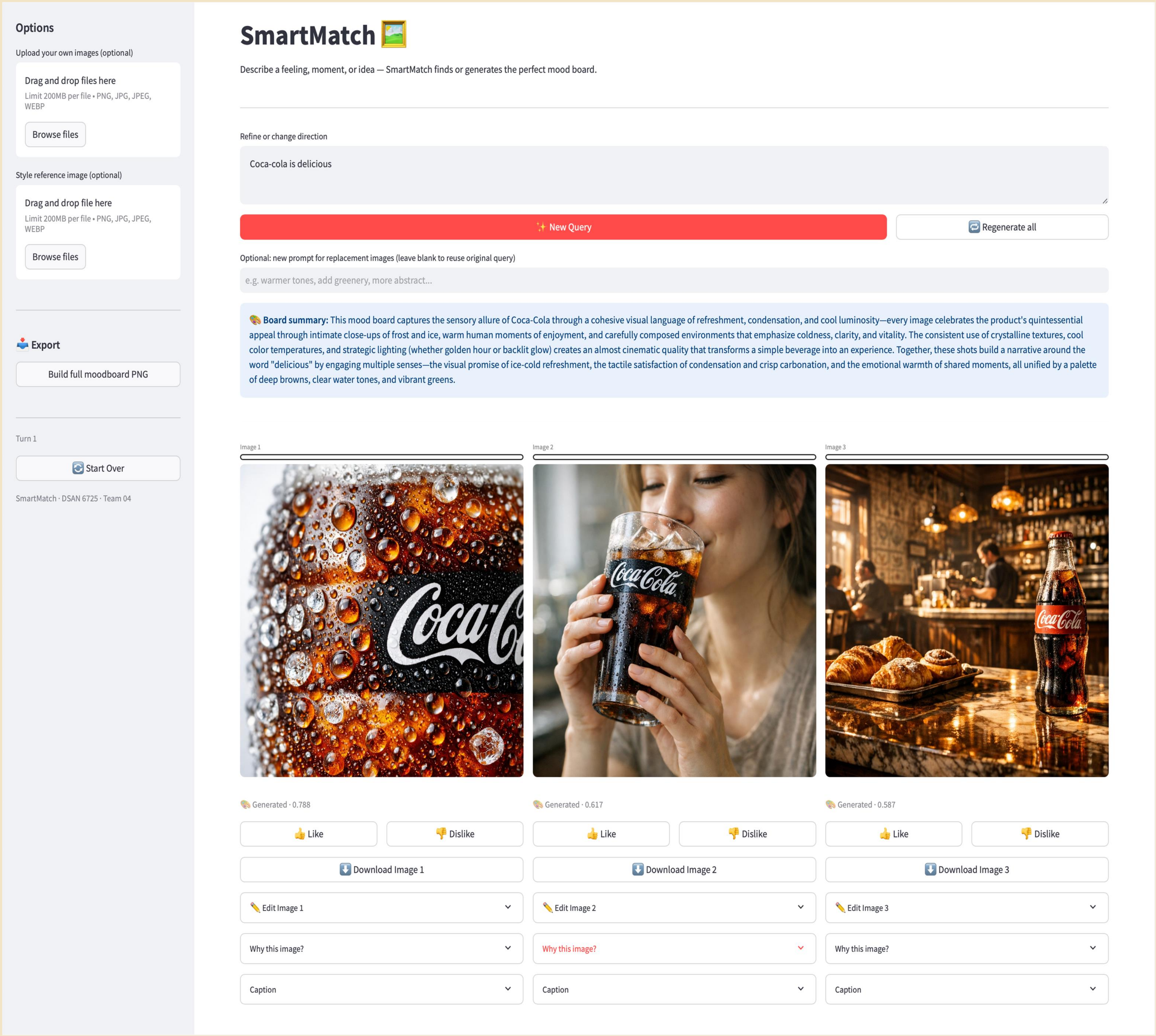
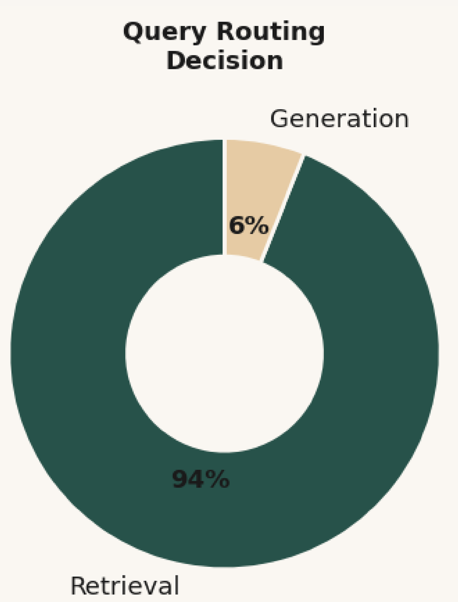


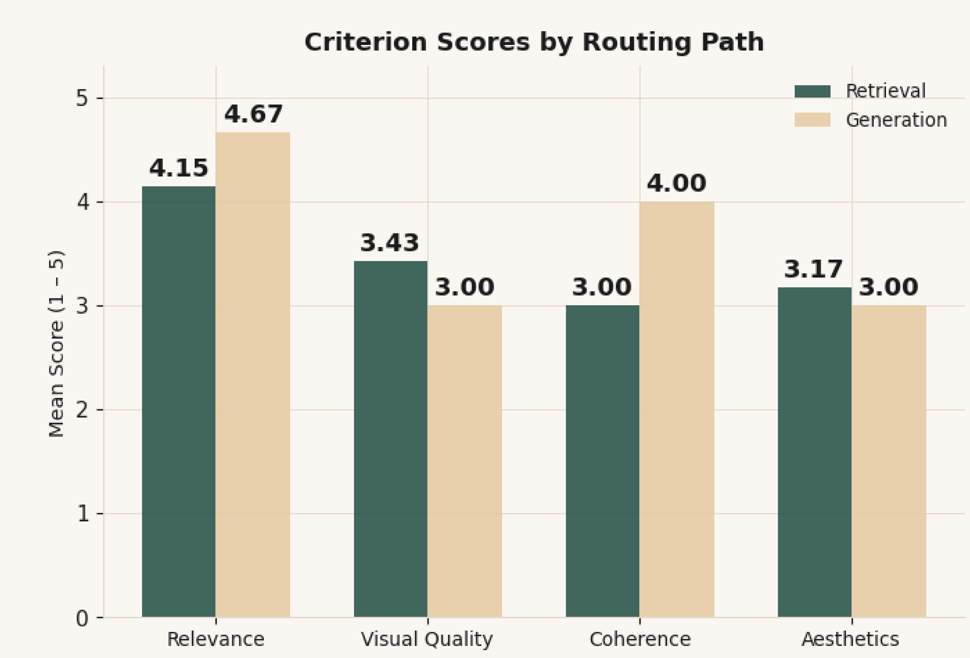
Fig. 1 SmartMatch System Architecture

- **Input Guardrail**: Blocks harmful queries.
- **Visual Concept Grounding Agent**: Decomposes user text into structured visual descriptors.
- **Hybrid Retrieval**: Retrieves candidates by fusing SigLIP-2 visual and per-field text embeddings.
- **Graph RAG Agent**: Deduplicates, expands, and re-ranks candidates.
- **Verification + Coherence Agent**: Validates each candidate and selects images for thematic and aesthetic balance.
- **Generation Fallback**: Diverse prompt synthesis when retrieval scores are low.
- **Justification Agent**: Generates per-image explanations and a board-level narrative grounded in captions and user intent.



Performance Evaluation





CONDITION	RELEVANCE	COHERENCE
A SigLIP-2 only, no grounding, no Graph RAG	2.8	2.6
B + LLM grounding +0.7	3.5	3.1
C + Graph RAG (dedup + expand + rerank) +0.6	3.8	3.7
D Full system (+ verification + coherence)	4.1	3.9

LLM-as-judge evaluation across 50 queries spanning five categories. Left: mean scores on four dimensions for mood boards from both routing path. Right: Incrementally adding pipeline components and reporting mean scores on two dimensions.

CONCLUSION

- LLM grounding raises retrieval relevance by decomposing abstract language into structured visual descriptors.
- Graph RAG contributes the coherence gain through connectivity-based reranking.
- Multi-turn refinement allows users to steer the board via like/dislike and chat feedback.
- Coherence remains the primary challenge, requiring diversity enforcement at the retrieval stage.